

MedTrack – Hospital Operations & Patient Analytics Dashboard Using Tableau

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Project: MedTrack – Hospital Operations & Patient Analytics Dashboard

Tool Used: Tableau Desktop

Dataset: Hospital Patient Dataset (100,000 Records)

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INTRODUCTION

Healthcare organizations generate massive amounts of patient and operational data every day. Managing admissions, monitoring hospital resources, evaluating department performance, and tracking patient flow manually is difficult and time-consuming.

The MedTrack Hospital Operations & Patient Analytics Dashboard was developed using Tableau to transform raw hospital data into interactive visual dashboards. The dashboard helps hospital administrators monitor hospital performance, improve operational efficiency, optimize resource utilization, and support data-driven decision making.

The project analyzes over 100,000 patient records and presents meaningful insights through interactive visualizations.

PROBLEM STATEMENT

Hospitals require an efficient system to monitor patient admissions, treatment performance, resource utilization, and department efficiency. Traditional reports are static and difficult to analyze.

The objective of this project is to build an interactive Tableau dashboard that enables users to:

- Monitor hospital performance
- Analyze patient flow
- Compare department efficiency
- Optimize hospital resources
- Improve operational decision-making

OBJECTIVES

The main objectives of the project are:

- Analyze hospital admission trends
- Monitor patient flow throughout the hospital
- Calculate important hospital KPIs
- Compare department performance
- Track readmission rates
- Analyze treatment costs
- Monitor bed utilization
- Evaluate occupancy rates
- Improve healthcare decision-making using interactive dashboards

DATASET DESCRIPTION

The dataset contains approximately **100,000 hospital patient records**.

Important Fields

- Patient ID
- Admission Date
- Discharge Date
- Department
- Diagnosis
- Severity Level
- Insurance Type
- Outcome
- Gender
- Age
- Average Length of Stay
- Wait Time
- Treatment Cost
- Readmission Rate
- Bed Utilization Rate
- Occupancy Rate

The dataset represents different departments including:

- Cardiology
- Neurology
- Oncology
- Emergency
- Orthopedics
- Pediatrics
- ICU
- General Surgery
- Dermatology
- Gynecology

TOOLS & TECHNOLOGIES

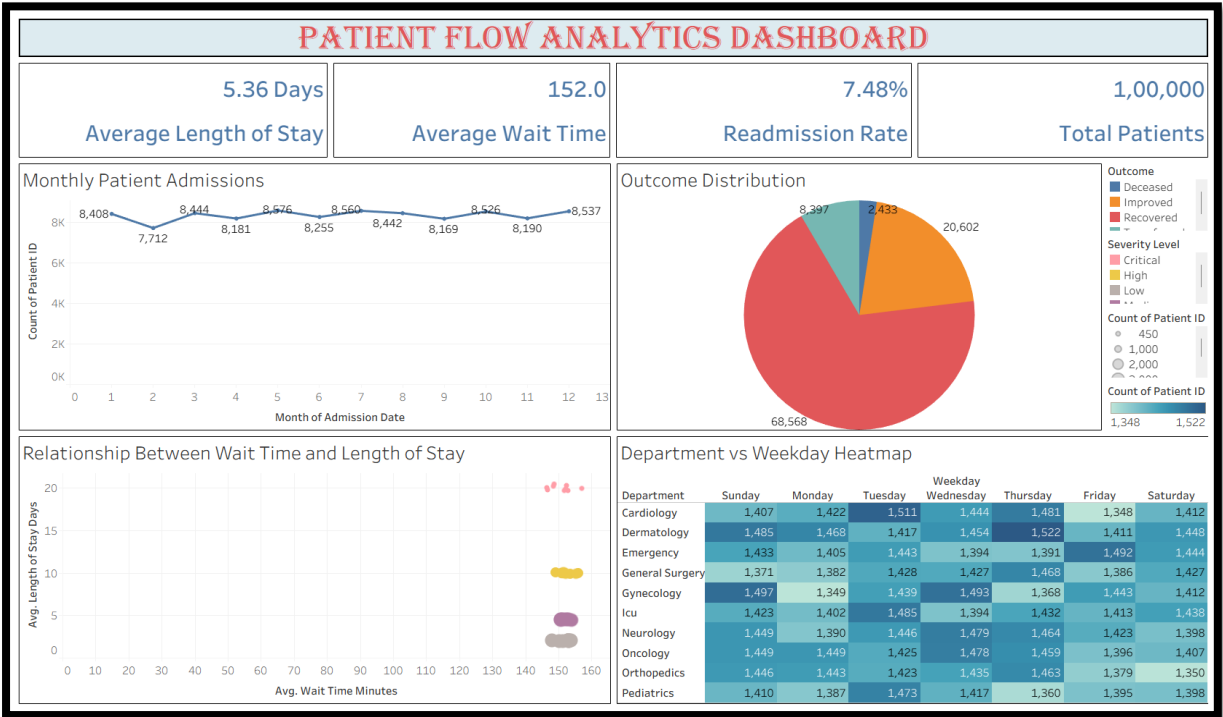
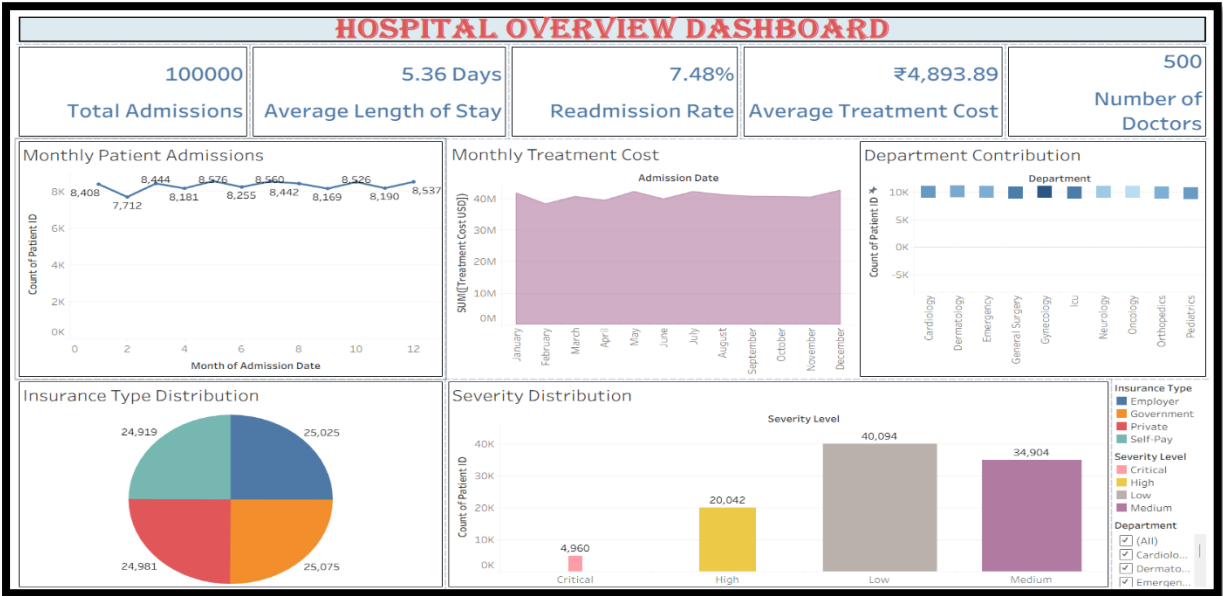
Component	Technology
Visualization Tool	Tableau Desktop
Dataset Format	CSV
Records	100,000
Charts	Line, Pie, Treemap, Heatmap, Bar, Scatter Plot, Area Chart
Dashboard Type	Interactive

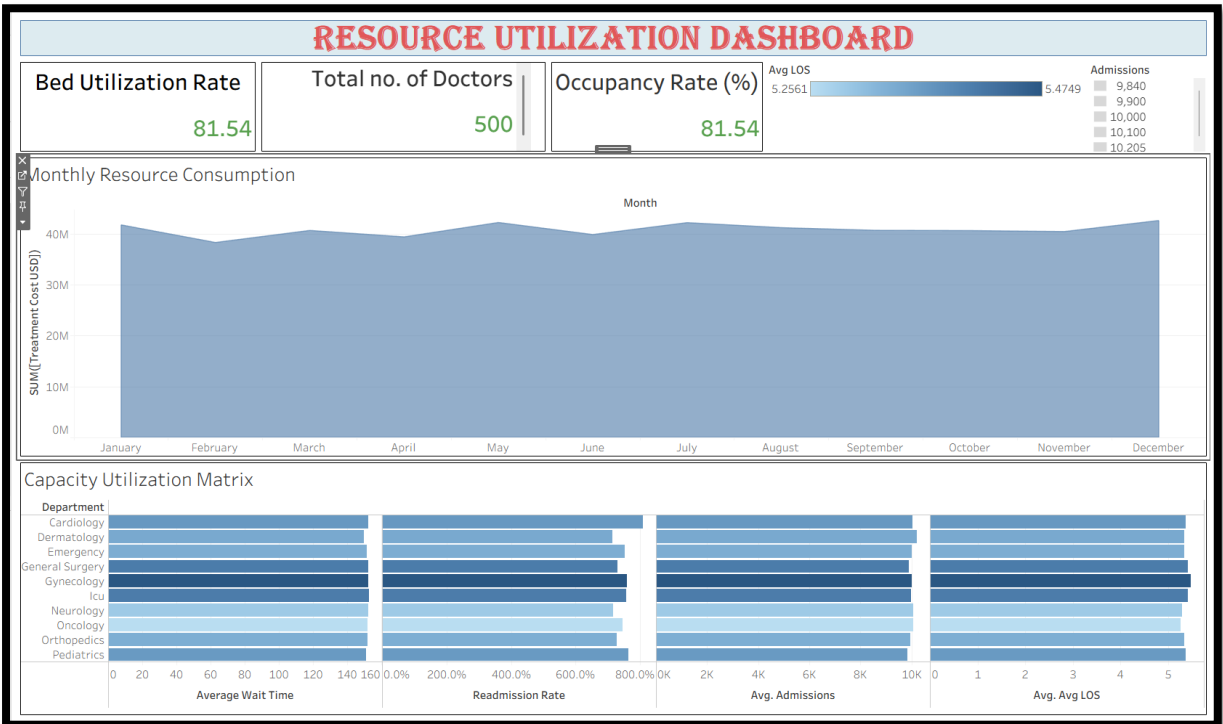
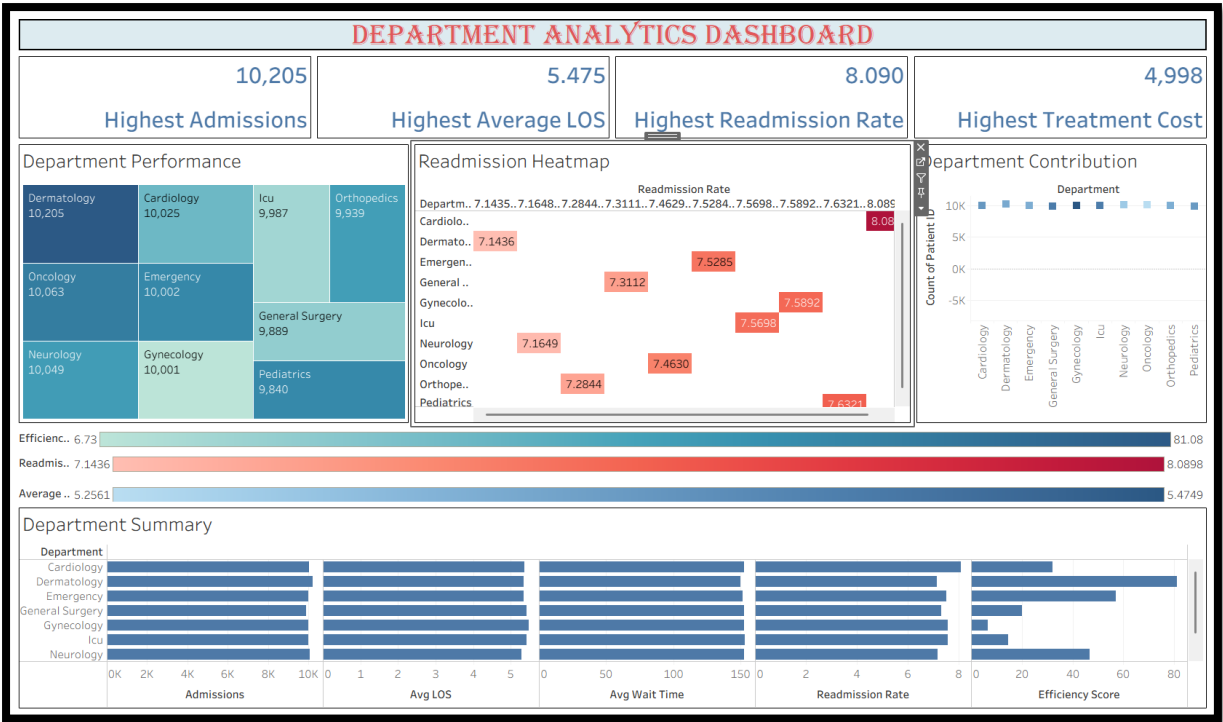
DATA PREPARATION

The following preprocessing steps were performed before visualization:

- Imported hospital dataset into Tableau
- Verified data types
- Converted Admission Date into Date format
- Created Month hierarchy
- Validated numerical measures
- Removed unnecessary fields
- Created calculated KPIs
- Checked missing values
- Organized dimensions and measures

DASHBOARD DEVELOPMENT





KPI DEFINITIONS

KPI	Description
Total Admissions	Total number of admitted patients
Average Length of Stay	Average hospitalization duration
Readmission Rate	Percentage of patients readmitted
Average Treatment Cost	Average treatment expense
Bed Utilization Rate	Percentage of beds utilized
Occupancy Rate	Percentage of occupied beds
Average Wait Time	Average patient waiting time
Number of Doctors	Total available doctors

TESTING AND VALIDATION

The dashboard was tested to ensure accurate calculations and proper interaction.

KPI Validation

All KPI values were cross-verified with the dataset.

Validated KPIs include:

- Total Admissions
- Average LOS
- Readmission Rate
- Bed Utilization Rate
- Occupancy Rate
- Treatment Cost

Result:

All KPI calculations were accurate.

Dashboard Interaction Testing

The following dashboard interactions were tested:

- Worksheet navigation
- Interactive filters
- Chart responsiveness
- Dashboard loading
- Tooltip display
- Cross-dashboard consistency

Result:

All dashboards functioned correctly.

Patient Flow Validation

Verified:

- Monthly admission trends
- Department-wise admissions
- Outcome distribution

- Wait time analysis
- Length of stay analysis

Result:

Patient flow metrics were validated successfully.

KEY INSIGHTS

The dashboard generated several important insights.

- Approximately 100,000 patient admissions were analyzed.
- Average Length of Stay is approximately 5.36 days.
- Readmission Rate is around 7.48%, indicating effective patient care.
- Bed Utilization Rate is approximately 81.54%, showing efficient use of hospital beds.
- Occupancy Rate remains around 81.54%.
- Average Treatment Cost is approximately ₹4,893.89.
- Most patients belong to Low and Medium severity categories.
- Department workload is relatively balanced across major specialties.
- Monthly admissions remain stable throughout the year.
- Resource utilization supports effective hospital capacity planning.

CONCLUSION

- The MedTrack Hospital Operations & Patient Analytics Dashboard successfully transforms large-scale hospital data into meaningful visual insights.
- The dashboard enables hospital administrators to monitor admissions, evaluate departmental performance, analyze patient flow, and optimize resource utilization. Interactive visualizations simplify decision-making and improve operational efficiency.
- This project demonstrates the practical application of Tableau for healthcare analytics and business intelligence.

FUTURE SCOPE

The dashboard can be further enhanced by integrating advanced analytics and real-time hospital systems.

Future improvements include:

- Real-time database integration
- AI-based patient admission prediction
- Bed allocation optimization using machine learning
- Patient risk prediction models
- Doctor scheduling optimization
- Emergency room analytics
- Mobile dashboard deployment
- Cloud-based healthcare reporting
- Predictive resource planning
- Integration with Hospital Management Systems (HMS)